

Techno-economic analysis of grid-integrated PV/wind systems for electricity reliability enhancement in Ethiopian industrial park

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ABSTRACT

In most developing countries, the electricity supply system is highly unreliable. Ethiopia is one of the least developed country in the world, and the existing distribution system of the country has encountered frequent power interruptions. During this interruption, diesel generator supplied the critical load of most industries in the country. This paper examines the feasibility of integrating PV/wind power systems into existing unreliable grid/diesel generator systems for supplying the critical loads of industrial parks in three different regions of Ethiopia. The study focused on how to provide a reliable supply with cost-effective and environment-friendly resources. Based on load variation, grid interruption, and meteorological data of the study areas, modeling and techno-economic analysis of grid-connected PV/wind/diesel systems were carried out using HOMER Pro software by exploring four different scenarios involving the consideration of unscheduled outages. Results showed that grid//diesel/PV/battery systems are technically, economically, and environmentally feasible for all three climate regions with the cost of energy at 0.044, 0.049, and 0.048 \$/kWh, respectively. Also found that excess electricity, cost of energy, and the net present cost is slightly increased with PV penetration, whereas the CO₂ emissions for these locations decreased by 45%, 44%, and 42% compared to the existing systems.

1. Introduction

Ethiopia is well endowed with various renewable energy resources, and has a potential to generate more than 60,000 MW of electric power from hydro, solar, wind, and geothermal sources (Federal Democratic Republic of Ethiopia, Ministry of water and energy, 2012; Ministry Of Water Irrigation and Electricity, 2017). Currently, the country has approximately 4284 MW of installed generation capacity to serve a population of about 105 million people (CIA The world fact book, 2017; Ministry Of Water Irrigation and Electricity, 2017). From the entire installed capacity, 89% of electricity is generated from hydropower (International Hydropower Association, 2018).

Even though the country has significant electricity generating potential, it is still characterized by being one of the least electrified countries of the world with frequent and prolonged grid outages. According to a report (World Bank Enterprise survey, 2015), in Ethiopia, there are 8.2 electrical outages in a typical month with an average outage duration lasting 5.8 h. All industrial, commercial and residential customers can be affected by this outage. Especially for industries, it is challenging to tolerate outages, since it causes much

revenue loss within half an hour of interruption.

To mitigate the negative impact of grid outage on the customer, in many locations around the globe, on-grid site was equipped with a diesel generator as a backup power source (Rodríguez-Gallegos et al., 2018). The majority of the industry zones in Ethiopia also use diesel generators as a backup (Carlsson et al., 2018). However, the combustion of diesel for power generation can produce undesirable gasses, like carbon dioxide (CO₂), carbon monoxide (CO), nitrogen oxides (NO_x), sulfur dioxide (SO₂), and particulate matter (PM) (Khalife et al., 2017). These gasses have a detrimental impact on the environment and future human life (Mohankumar and Senthilkumar, 2017; Wei and Geng, 2016).

Based on the continuing commercial maturation of distributed generation (DG) technologies like PV and wind turbines, the addition of these renewable energy sources into diesel generator alleviates environmental concern and fuel price uncertainties (Hossain et al., 2017; Ramli et al., 2018). Ethiopia has also a target to reach a share of 15–20 % of its energy supply from non-hydropower-based renewable resources (solar, wind, and geothermal) by 2020 to realize a zero net carbon emission up to 2025 (Federal Democratic Republic of Ethiopia,

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Nomenclature

COE	Cost of energy
CRF	Capital recovery factor
CHP	Combined heat and power
DG	Distributed generation
EEU	Ethiopian electric utility
GTP	Growth and transformation plan
HOMER	Hybrid optimization model for electric renewables
IP	Industrial park
NASA	National aeronautics and space administration
NPC	Net present cost
NMA	National meteorology agency
PV	photovoltaic
RF	Renewable fraction
SNNP	Southern nation nationalities and people
C_p	The power coefficient.
C_T	The total annualized cost (\$/year)
E_b	The battery energy in time interval
E_{gs}	Energy sold to the grid per year (kWh/yr)
F_1	The fuel curve slope in (L/hr/kW _{output})
f	Inflation rate (%).
F	Fuel consumption rate (L/hour)
F_0	Generator fuel curve intercept coefficient
f_{pv}	The PV derating factor
G_{SC}	The solar constant = 1367 (W/m ²)
G_T	The solar radiation incident on the PV array in the current time step (kW/m ²)
$G_{T,STC}$	The incident radiation at standard test conditions (1kW/m ²)

H	Monthly average daily radiation on a horizontal surface
H_0	Monthly average daily extraterrestrial radiation on a horizontal surface (MJ/m ²)
i	The annual real interest rate (%),
i_n	The nominal interest rate (%)
N	The lifetime of the project (year)
n_{bb}	The battery charging efficiency
n_{bi}	The inverter efficiency
P_{gen}	The electrical output of the generator
P_{pv}	The average power output of the PV array
S	Monthly average daily number of hours of bright sunshine
S_0	Maximum possible daily bright sunshine hour
T_c	The PV cell temperature in the current time step (°C)
$T_{c,STC}$	The PV cell temperature under standard test (25 °C)
$v(z)$	Wind speed at height of Z m above ground (m/s)
V_r	wind speed at reference height (m/s)
Y_{gen}	The rated capacity of the generator (kW)
Y_{pv}	The peak power output of PV
Z	Elevation above ground (m)
Z_0	Surface roughness length (m)
Z_r	Reference height (m)
A_T	the swept area (m ²)
α_p	The temperature coefficient of PV (%/°C)
σ	The self-discharging factor
ρ	Air density (kg/m ³)
δ	Declination angle (°)
ϕ	Latitude angle (°)
ω_s	The sunset hour angle (°)

2013; World Bank, 2018). Consequently, the government of Ethiopia is motivating on the exploration of renewable distributed generation technologies.

To achieve the targeted goals of the country, analyzing grid-connected solar/wind power systems could be one of the viable options. Such kind of grid-tied renewable distributed generation systems not only offer environmental benefit and electric service, but also provides customer benefits through selling excess power generated by the systems to the grid (International renewable energy agency (IRENA), 2018; Numbi and Malinga, 2017). Various studies have analyzed the techno-economic feasibility of grid-integrated renewable distributed generation systems using different optimization techniques.

Islam (2018) presented the techno-economic analysis of grid-tied PV/hydropower systems for a huge office building in the southeastern part of France using HOMER software. Results showed that PV/grid systems were more cost effective than grid-only systems. In Ref. Rehman et al. (2017), the economic and technical analysis of grid-connected PV systems at different sites of Saud Arabia was explored using RETScreen simulation tool. The results indicated that from selected site Bisha was the best area for the implementation of grid-connected solar PV systems. Gonzalez et al. (2018) discussed the economic and environmental impact of grid-connected PV/wind/biomass systems in central Catalonia using genetic algorithm optimization technique. Results showed that the grid-integrated hybrid renewable systems were less cost-effective than the grid-only systems, but it has lower environmental impacts compared to grid-only systems. Li et al. (2018) evaluated the techno-economic performance of grid-connected PV systems in the five climate zones of China using HOMER software. The results indicated that grid/PV systems were technically and economically feasible for all five climate zones. Kazem et al. (2017) studied the techno-economic viability of 1 MW PV grid-connected systems in Oman with the help of numerical simulation using MATLAB developed code. They found that the proposed grid-integrated PV systems were

economically feasible. Bastholm and Fiedler (2018) evaluated the techno-economic analysis of grid-connected PV/diesel/battery power systems under national grid blackouts in Tanzania using HOMER simulation tools. From the result, they found that for power outages above 0.75 h per day, grid-tied PV/diesel/battery systems were economically feasible for the considered load in the locations. Al Garni et al. (2018) analyzed grid-tied PV systems under different tracking condition in Makkah, Saudi Arabia using HOMER simulation software. The results suggested that grid-connected vertical axis tracker was more cost effective than horizontal axis tracker for the considered location. Rehman et al. (2018) analyzed grid-connected PV systems for household in Pakistan under grid outage consideration using HOMER Pro simulation tool. The results pointed that grid-tied PV systems with battery storage were both economically and technically viable for the case under consideration. Jahangiri et al. (2019) evaluated the techno-economic and environmental analysis of grid/wind/ PV systems for the provision of electricity and hydrogen in the southern location of Iran using HOMER Pro software. The results depicted that grid/wind/PV systems were the most optimal mode for electricity generation with COE 1.55\$/kWh, and 0.496 \$ per kg of hydrogen generation price was achieved using the main grid only. Mukisa et al. (2019) studied the feasibility assessment of grid-integrated rooftop solar photovoltaic systems for 36 industrial application in Uganda. Google Earth and Azimuth tools were used to determine the rooftop area. The results showed that the proposed PV systems were economically viable.

Other studies have also addressed the techno-economic analysis of off-grid hybrid power systems in the combination of PV/wind/battery (Fulzele and Diagavane, 2018; Haratian et al., 2018; Sanajaoba Singh and Fernandez, 2018; Zhang et al., 2018), PV/diesel/battery (Das and Zaman, 2019; Fodhil et al., 2019; Halabi et al., 2017), PV/wind/battery/diesel (Mandal et al., 2018; Olatomiwa et al., 2015a; Olatomiwa et al., 2015b; Olatomiwa et al., 2018; Razmjoo et al., 2019; Tawfik et al., 2018), PV/fuel cell/diesel (Jamshidi and Askarzadeh, 2019), and

PV/wind/diesel systems (Shezan et al., 2017; Zhou et al., 2018) in different locations.

As it is mentioned in the above literature, the techno-economic analysis of off-grid and grid-connected hybrid renewable systems have been investigated in several studies. However, the techno-economic comparative study of grid-connected renewable DG by considering the effect of an outage is not explored thoroughly enough. Notably, the effect of unscheduled outages has so far not been considered in sufficient detail. Therefore, the main focus of this study is to examine the techno-economic performance of grid-connected PV/wind power systems under unscheduled grid outage consideration for the industrial park (IP) load, in three different regions of Ethiopia. Previously diesel generator was used as a backup to support the unreliable hydro-based grid power of the IP. In this regard, this study also investigated the means to reduce the consumption of diesel fuel and the working hours of diesel generators. The performance of the designed system is simulated using the HOMER Pro optimization software (Das et al., 2017; Halabi and Mekhilef, 2018).

The rest of the paper is structured as follows: Section 2 presents the site description of the study regions. Methodology and modeling of grid-connected PV/wind power systems of the locations are introduced in Section 3. Simulation results and discussions are provided in Section 4. Finally, the conclusions of this study are described in Section 5.

2. Background information

2.1. Site background

The second Growth and Transformation Plan (GTP II) of Ethiopia was endorsed by the council of ministers to guide development endeavors within the country under the implementation for 2015–2020 (Federal Democratic Republic of Ethiopia, National planning

commission, 2016). Development and expansion of industrial parks are one of the sectors to which GTP II gave attention, and based on the plan several industrial parks are being developed throughout the country (Embassy of Federal Democratic Republic of Ethiopia in Japan, Tokyo, 2018; United Nation Development Organization, 2018). However, due to the high power demand and the climatic impacts on the hydropower system, the industrial park has been facing some power outages that affect the manufacturing process.

Hawassa IP, Kombolcha IP, and Adama IP were the three main industrial parks in the country, and currently unscheduled grid outages shown in Table 2 are considered. To analyze the potential of renewable energy (solar/wind) and to design a reliable supply, these three region's representative parks were also chosen under this study. The geographical layout, overview, and coordinates of the locations are shown in Fig. 1 and Table 1, respectively.

3. Methodology

The feasibility, techno-economic, optimization, and sensitivity analysis of the grid-connected PV/wind power systems for the selected locations are carried out using HOMER Pro, with the climate, load, grid outage, and economic data as inputs as shown in Fig. 2.

3.1. Resource assessment

In the following subsection, the solar radiation potential and wind resources of the locations are analyzed.

3.1.1. Solar irradiation

For many developing countries including the investigated sites, there is no properly recorded solar radiation data. What usually available is sunshine duration data. However, it is possible to compute the

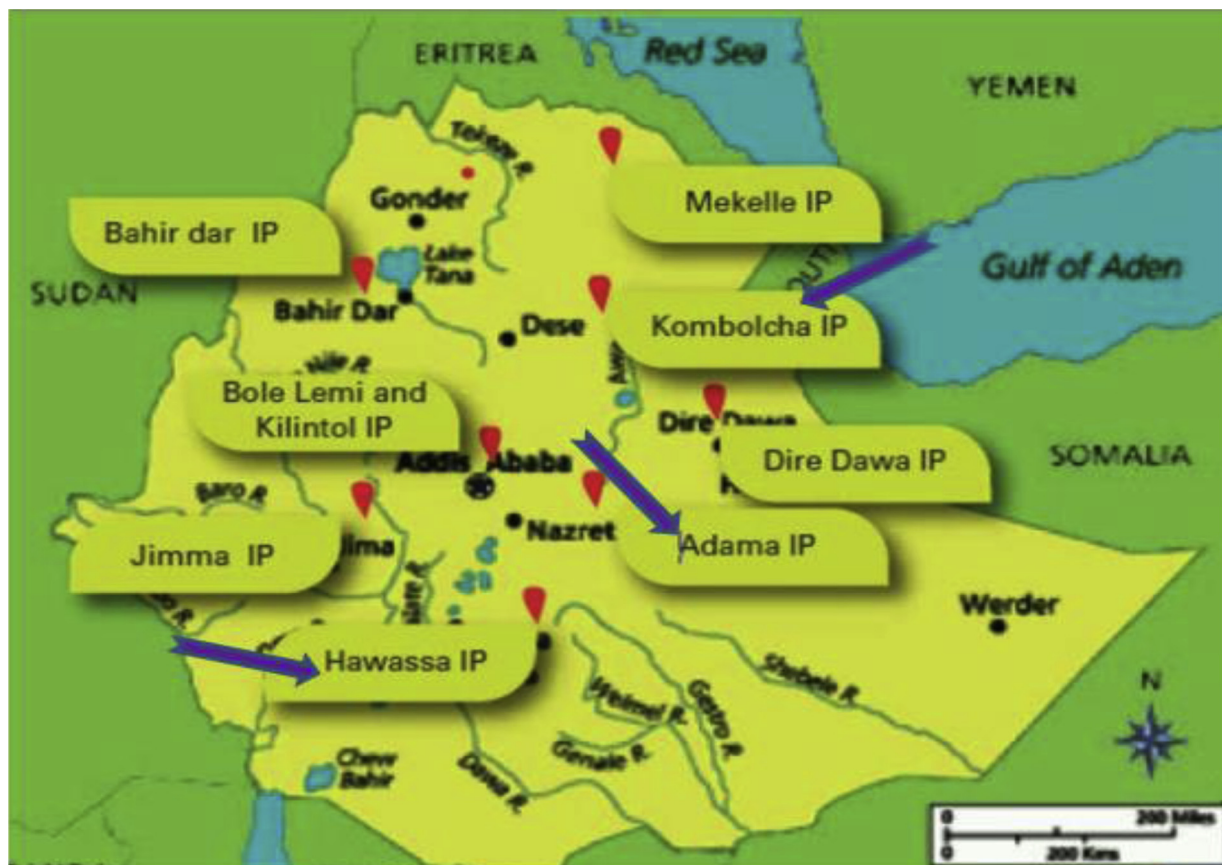


Fig. 1. Geographical layout of the project area (Ethiopian investment commission, 2017a), Google Map 2018 edited].

Table 1

Overview and geographical coordinate of the study region (United Nation Development Organization, 2018; Fazzini et al., 2015; Gonfa, 1996; Ethiopian investment commission, 2017b).

Name of the IP	Region	Latitude (°N)	Longitude (°E)	Elevation (m)	Climate regions	Land size of the park in hectare	production	Incoming Voltage to IP substation
Hawassa IP	SNNP	7.06	38.45	1,694	Tropical rainy climate (Savanna)	300	Garment	132kV
Kombolcha IP	Amhara	11.08	39.73	1,857	Warm temperature rainy climate	700	Garment and Food	132kV
Adama IP	Oromia	8.54	39.26	1,622	Tropical rainy climate	2,000	Textile and Food	132kV

Table 2

Yearly average interruption of the industrial parks.

Name of feeder	Year	Average interruption frequency (Int/year)	Average interruption duration (hr/year)
Hawassa IP	2018	61	92
Kombolcha IP	2018	94	236
Adama IP	2018	89	167

average monthly solar radiation data using empirical Eq. (1) (Bekele and Palm, 2010; Duffie and Beckman, 2006; Mulugeta and Drake, 1996).

$$H = H_o \left(a + b \left(\frac{S}{S_o} \right) \right) \quad (1)$$

Where H is the monthly average daily radiation on a horizontal surface (MJ/m^2), H_o is the monthly average daily extraterrestrial radiation on a

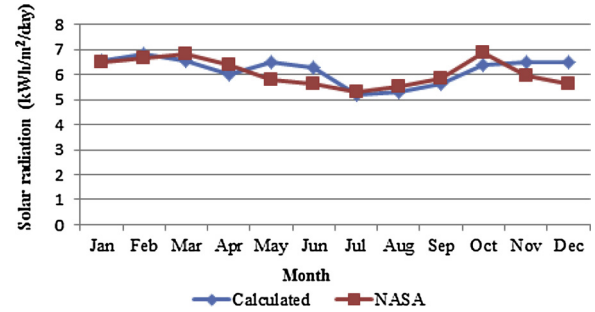


Fig. 3. Monthly average calculated and NASA solar radiation data for Hawassa IP.

horizontal surface (MJ/m^2), S is the monthly average daily number of bright sunshine hours, and S_o is the maximum possible daily hours of bright sunshine. Coefficients a and b are regression coefficients based

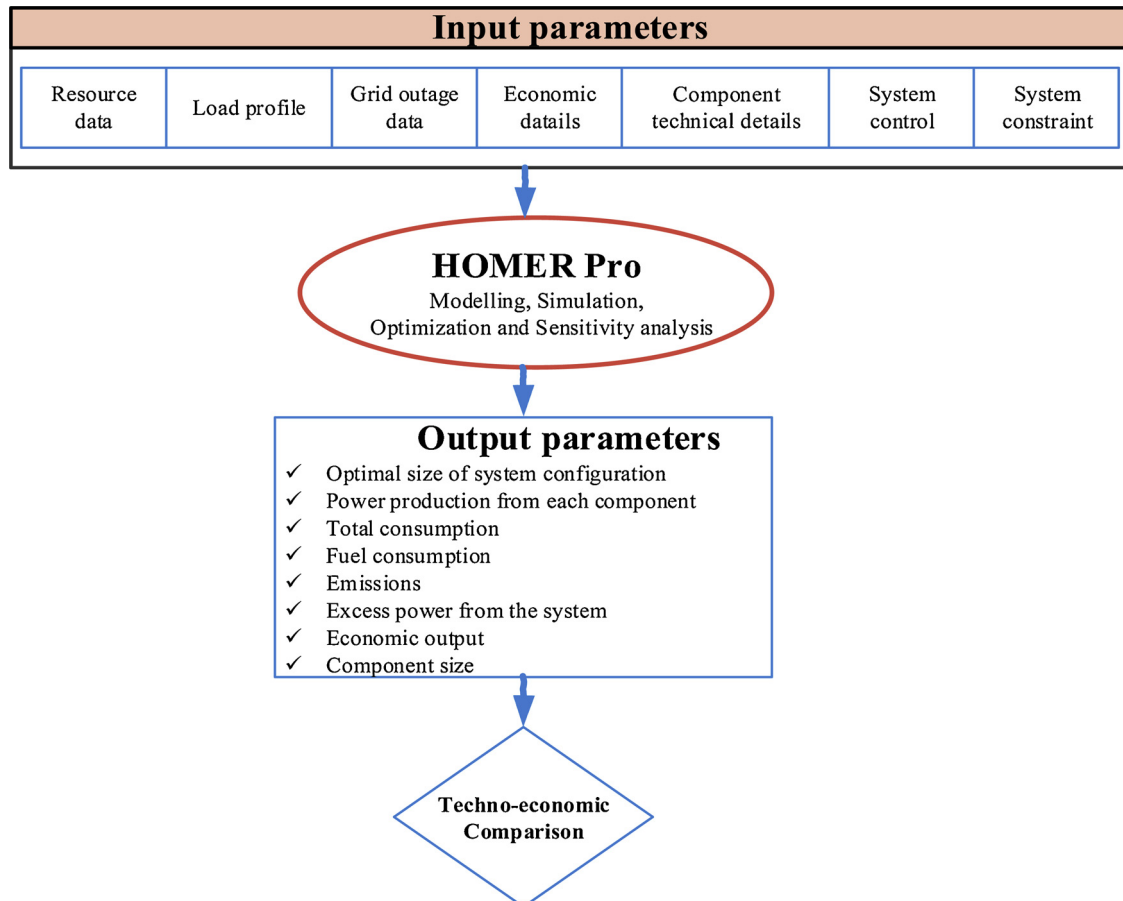


Fig. 2. Methodological steps for the study.

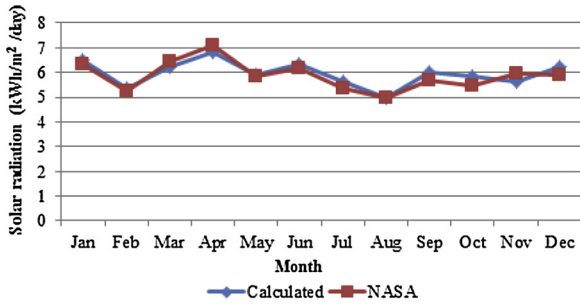


Fig. 4. Monthly average calculated and NASA solar radiation data for Kombolcha IP.

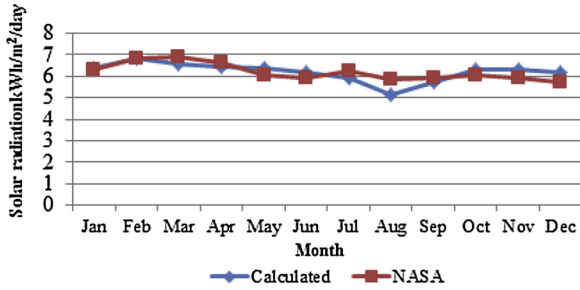


Fig. 5. Monthly average calculated and NASA solar radiation data for Adama IP.

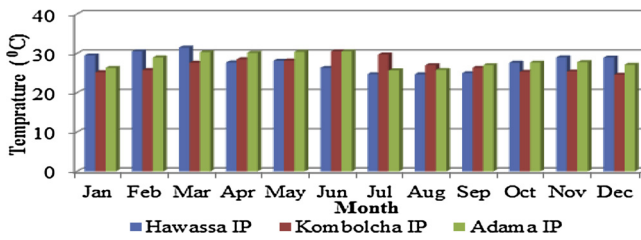


Fig. 6. Monthly average ambient temperature data for the study sites.

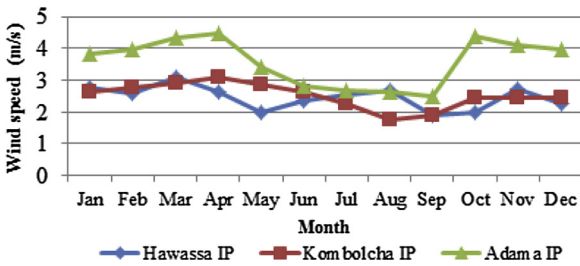


Fig. 7. Monthly average wind speed at 10 m height.

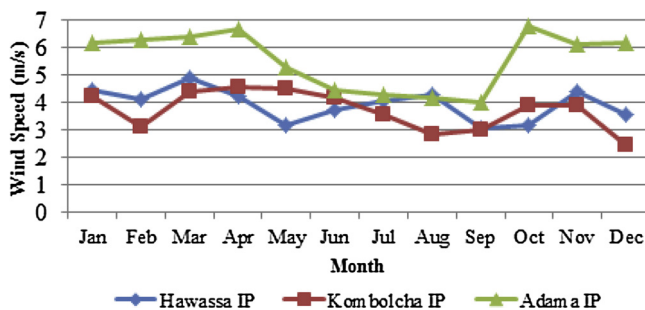


Fig. 8. Monthly average wind speed at 50 m height.

on the coordinates of a location. In this study, the regression coefficients having average values of $a = 0.33$ and $b = 0.43$ were used (Mulugeta and Drake, 1996). The Monthly average extraterrestrial radiation is computed using Eq. (2).

$$H_o = \frac{24 \cdot 3600 \cdot G_{SC}}{\pi} \left(1 + 0.033 \cdot \cos \left(\frac{360 n_d}{365} \right) \right) \cdot \left(\cos \phi \cos \delta \sin \omega_s + \frac{\pi \omega_s}{180} \sin \phi \sin \delta \right) \quad (2)$$

Where n_d is the day number starting from January 1st as 1, G_{SC} is the solar constant (1367 W/m^2), ϕ is the latitude of the location ($7^\circ 04' 0.66''\text{N}$), δ is the declination angle ($^\circ$), given by Eq. (3), and ω_s is the sunset hour angle ($^\circ$), which is given by Eq. (4) below.

$$\delta = 23.45 \sin \left(360 \frac{284 + n_d}{365} \right) \quad (3)$$

$$\omega_s = \cos^{-1}(-\tan \phi \tan \delta) \quad (4)$$

The maximum possible sunshine duration S_o is given by;

$$S_o = \frac{2}{15} \cos^{-1}(-\tan \phi \tan \delta) \quad (5)$$

Based on the above equations, solar radiation data of the locations were calculated. The computed annual average solar radiation for Hawassa IP, Kombolcha IP, and Adama IP were $6.18 \text{ kWh/m}^2/\text{day}$, $5.96 \text{ kWh/m}^2/\text{day}$ and $6.20 \text{ kWh/m}^2/\text{day}$ respectively. The yearly average solar radiation of Hawassa IP and Adama IP is nearly the same because both of them are located in the same climate zone. Because of warm temperature and rainy climate nature, the average annual radiation of Kombolcha IP is slightly lower than that of the two sites.

NASA database (NASA Prediction of worldwide energy resource, 2018) was additionally used to validate the calculated result. The annual average radiation data taken from NASA for Hawassa IP, Kombolcha IP, and Adama IP were $6.07 \text{ kWh/m}^2/\text{day}$, $5.87 \text{ kWh/m}^2/\text{day}$, and $6.15 \text{ kWh/m}^2/\text{day}$ respectively. The monthly calculated solar radiation data and the data obtained from NASA for each region are shown in Figs. 3–5. For this investigation, the calculated data is considered.

In order to determine the actual output power from solar PV, knowing the ambient temperature of the sites is very important. Fig. 6 below shows the average monthly temperature of the sites taken from the National Meteorology Agency (NMA) of Ethiopia. The annual average temperatures in Hawassa IP, Kombolcha IP, and Adama IP were 27.9°C , 27°C , and 28.14°C , respectively.

3.1.2. Wind resource assessment

Wind speed data of the locations were obtained from National Meteorological Agency (NMA). The NMA collects wind speed data at 2 m height, which is below the reference standard of 10 m height placed by the World Meteorological Organization (WMO). However, if the wind speed measurement was not made at the reference height, it is possible to extrapolate to the selected wind turbine hub height using Eq. (6) (Drake and Mulugetta, 1996; Manwell et al., 2009).

$$V(z) \ln \left(\frac{Z_r}{Z_o} \right) = V(Z_r) \ln \left(\frac{Z}{Z_o} \right) \quad (6)$$

Where Z_r is the reference height (m), Z is the height for which wind speed is to be determined (m), Z_o is the surface roughness (0.1–0.25 for crop land), $V(z)$ is the wind speed at height of z m (m/s), and $V(Z_r)$ is the wind speed at the reference height (m/s). Based on the location of the site, the value of Z_o has been taken as 0.2.

The extrapolated annual wind speed data obtained from the NMA at 10 m height for Hawassa IP, Kombolcha IP, and Adama IP were 1.65, 1.72, and 2.85 m/s, respectively.

The estimated annual average wind speed data taken from the NASA data base for the three sites were 2.48, 2.5, and 3.57 m/s at 10 m height

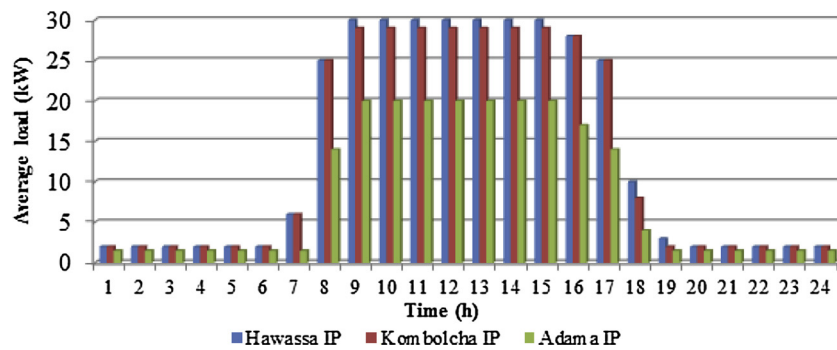


Fig. 9. The average daily critical load profile of the parks (year 2018).

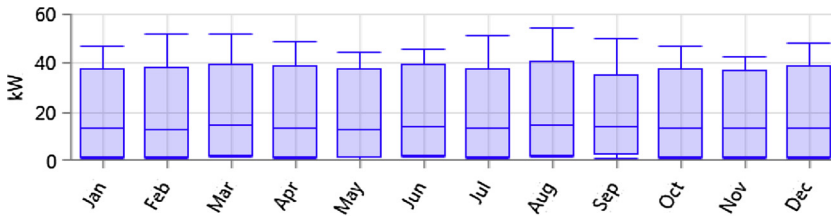


Fig. 10. Monthly average critical load profile of Hawassa IP (year 2018).

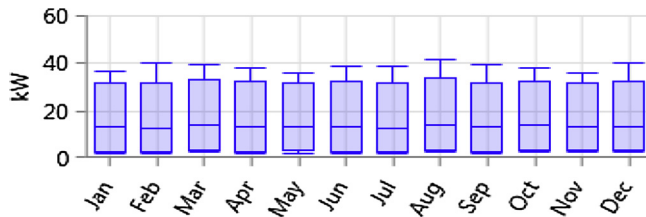


Fig. 11. Monthly average critical load profile of Kombolcha IP (year 2018).

(NASA Prediction of worldwide energy resource, 2018). These wind speed values are relatively high compared with the NMA data. In general, measurements at 2 m height are error prone due to different kinds of shading and vegetation. Therefore, the data obtained from NASA were considered for this study. The probable monthly average wind speed for the locations at 10 m and 50 m are given in Figs. 7 and 8, respectively.

3.2. Load profile of the location

The total load profiles for the sites were obtained from the respective regions of Ethiopian Electric Utility (EEU) for the year 2018. According to the collected load data, the estimated average daily electricity consumption of the parks was 72 MWh, 43 MWh and 38 MWh. Out of this load an average of 329 kWh/day, 321 kWh/day, and 208 kWh/day consumptions were considered as a critical load, as shown in Fig. 9. The unreliable grid and standby diesel generator supplied these critical loads. The remaining loads were supplied only by

an unreliable grid. The proposed grid-tied PV/Wind power systems were also modeled by considering the critical load of the sites only.

Based on their product size, the load profiles of the three IPs are different. Peak load demand occurred from 8:00 to 17:00 h in the day. This was mainly because of all the industrial park had started their manufacturing process. The monthly average critical load profiles are shown in Figs. 10–12.

3.3. Grid outage

In order to design a reliable system with the optimum size of distributed generation, knowing the exact value of interruption data is very important. During this investigation, the line interruption duration and interruption frequency of the sites for the year of 2018 have been collected from EEU of each region. The collected data are shown in Table 2.

3.3.1. Electricity tariff in the country

The current average electricity tariff is around 0.04 \$/kWh, which is one of the lowest cost of energy in Africa (Ethiopian Electric utility current tariffs, 2019). So far, the tariff in the country is subsidized by the government.

3.4. Description of the grid-connected PV/Wind power systems

The proposed scheme set out in Fig. 13 foresees the addition of solar PV, wind turbine, battery bank, and converters to support the diesel generators during outage and to increase the grid sales during normal

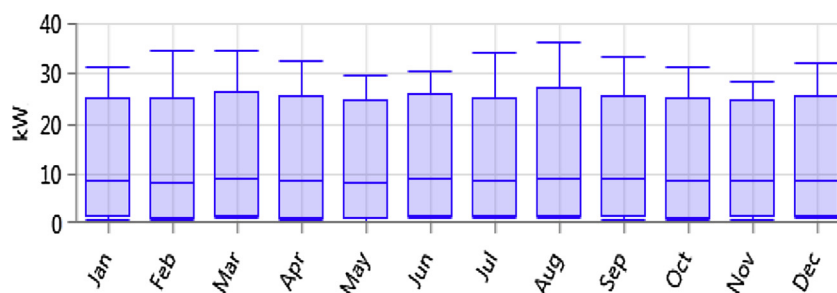


Fig. 12. Monthly average critical load profile of Adama IP (Year 2018).

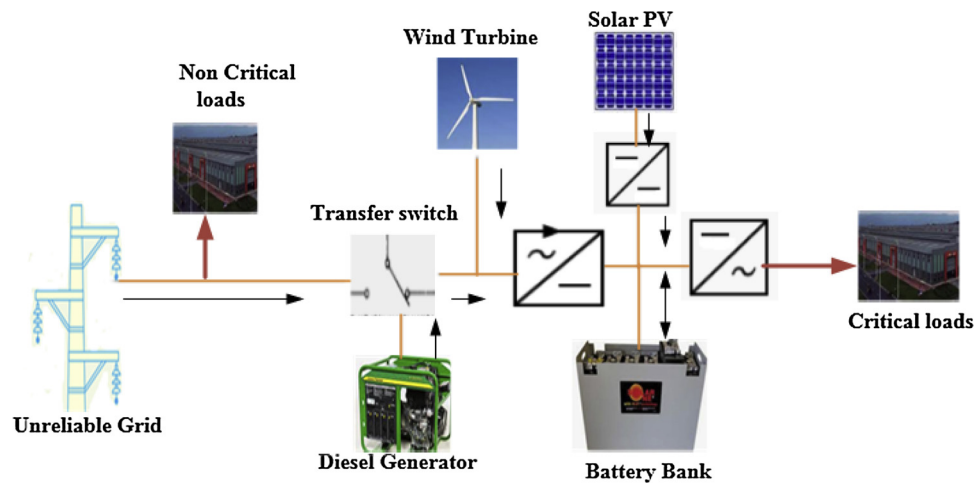


Fig. 13. General scheme of the proposed grid-tied PV/Wind systems.

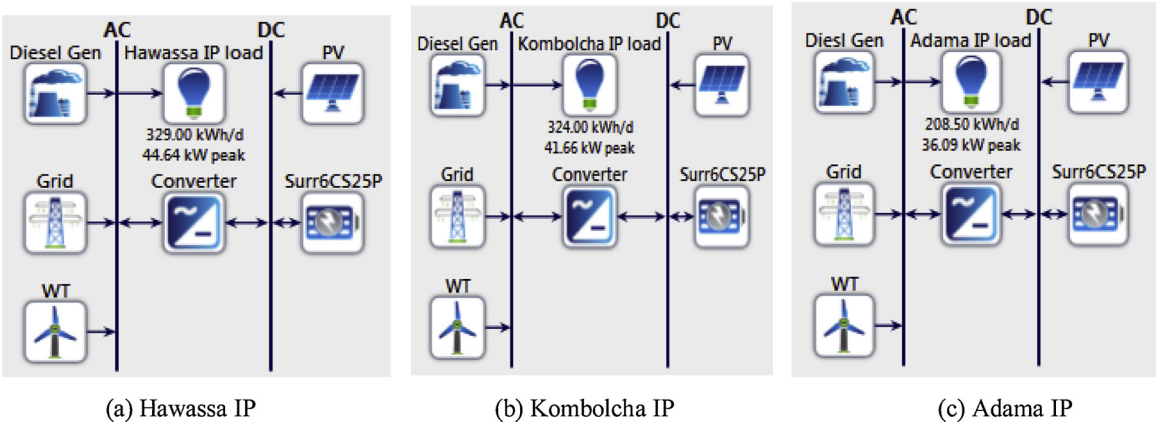


Fig. 14. Schematic diagram of grid-connected power system in HOMER Pro for the study regions.

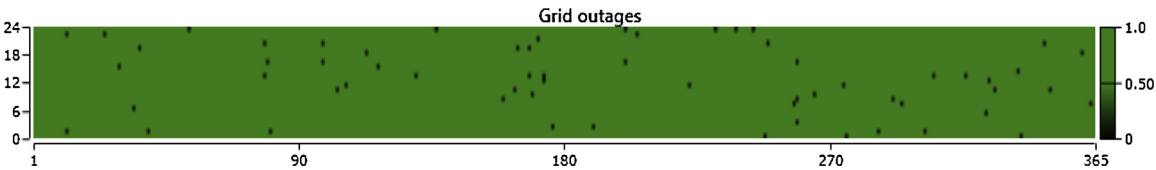


Fig. 15. Random grid outages in HOMER for Hawassa IP.

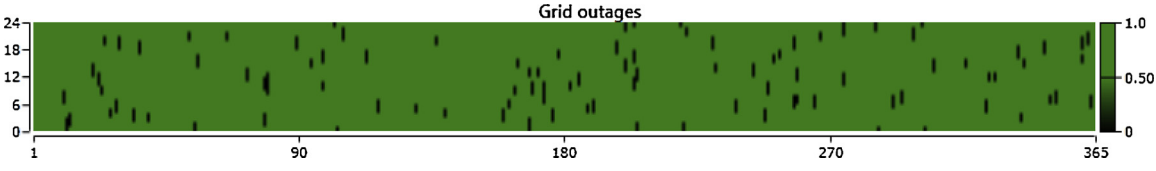


Fig. 16. Random grid outages in HOMER for Kombolcha IP.

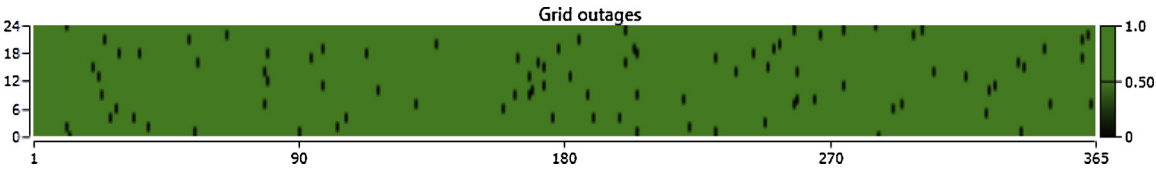


Fig. 17. Random grid outages in HOMER for Adama IP.

Table 3
Solar PV module technical and economic data.

Parameter	Specification
Panel type	Flat plate
Operating temperature	47 °C
Temperature coefficient of power	− 0.40 °C
Nominal operating cell temperature	17.02%
Derating factor	0.85
Capital cost	800 \$/kW
Operation and maintenance cost	8 \$/kW/yr.
Lifetime	25 years

operation. The representation of Fig. 13, after modelled by HOMER Pro for the study regions, is shown in Fig. 14.

3.4.1. Grid modeling

HOMER Pro allows to model unreliable grid by considering scheduled and unscheduled outages. In this study, an unreliable grid model with unscheduled grid outage was considered. In order to model an unreliable grid with unscheduled outage, mean failure frequency, mean repair time, and variability in repair time are given as an input. Based on the given input and by selecting a pseudo-random time step from a full year simulation period, random outages were generated by the software. Unscheduled grid outages imported from the software simulations are shown in Figs. 15–17. From the figures, the black spot indicates the random outage throughout the year and the regular grid operation appears in a green. During these outages, the distributed generators are forced to be on.

3.4.2. Solar photovoltaics

The PV panel is modeled as a component that generates DC power in direct proportion to the solar radiation. The output power obtained from the PV module is given by the following equation (Ismail et al., 2013; Soshinskaya et al., 2014; Thomas et al., 2016):

$$P_{pv} = Y_{pv} f_{pv} \left(\frac{G_T}{G_{T,STC}} \right) [1 + \alpha_p (T_c - T_{c,STC})] \quad (7)$$

Where: Y_{pv} is the peak power output [kW], f_{pv} is the PV derating factor, G_T is the solar radiation incident on the PV array in the current time step [kW/m^2], $G_{T,STC}$ is the incident radiation at standard test conditions [$1 \text{ kW}/\text{m}^2$], α_p is the temperature coefficient of power [$\%/^\circ\text{C}$], T_c is the PV cell temperature in the current time step [$^\circ\text{C}$], and $T_{c,STC}$ is the PV cell temperature under standard test conditions [25°C].

If the effect of temperature on the PV array is not considered, the temperature coefficient of power is zero; thus, the above equation is simplified into Eq. (8):

$$P_{pv} = Y_{pv} f_{pv} \left(\frac{\tilde{G}_T}{\tilde{G}_{T,STC}} \right) \quad (8)$$

In this investigation, the effect of temperature is taken into account. The specifications for the selected PV panel in this study are given in Table 3. The capital cost of solar PV is given based on the current local

Table 4
Technical and cost data of the selected wind turbine.

Parameter	Specification
Model	Bergey XL10
Rated power	10 kW
Cut-in wind speed	3 m/s
Cut-off wind speed	20 m/s
Hub height	30 m
Capital cost /unit	35,000 \$
Replacement cost /unit	30,000 \$
Operation and maintenance cost/unit	1,000 \$/year
Lifetime	20 years

Table 5
Technical and economic data of the selected battery.

Parameter	Specification
Model	Surrette 6CS25P
Nominal voltage	6V
Maximum capacity	1156 Ah
Minimum state of charge	40%
Roundtrip efficiency	80%
Lifetime throughput	9645Ah
Capital cost	900 \$
Replacement cost	700 \$
Operation and maintenance cost	9 \$/year

Table 6
Cost and technical data of the converter.

Parameter	Specification
Capital cost	650 \$
Replacement cost	600 \$
Operation and maintenance cost	6 \$/year
Efficiency	95%
lifetime	15 years

Table 7
Technical and economic data of considered diesel generator in three industry parks.

Parameter	Specification
Generator type	Synchronous
Rated power	50 kW, 50 kW, 48kW
Capital cost	0
Operation and maintenance cost	0.5 \$/hour
Diesel price in the country	0.62 \$/liter
lifetime	25,000 hours

price in the country.

3.4.3. Wind turbine

A wind turbine operates by converting the kinetic energy in the

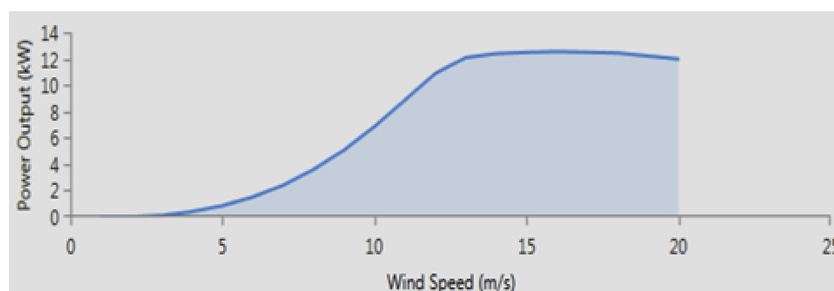


Fig. 18. The power-speed characteristic curve of Bergey 10 kW wind turbine.

Table 8
Monthly average electric production of grid/diesel systems in the three study regions.

Month	Hawassa IP		Kombolcha IP		Adama IP	
	Grid (kW)	Diesel (kW)	Grid (kW)	Diesel (kW)	Grid (kW)	Diesel (kW)
Jan	33.20	3.43	32.41	5.11	24.21	4.35
Feb	33.13	3.31	32.31	4.58	25.03	3.56
Mar	33.45	3.89	32.48	4.79	25.50	3.23
Apr	33.14	3.59	32.53	5.98	25.21	4.21
May	33.00	1.98	32.52	4.43	24.92	3.10
Jun	33.23	4.98	32.80	5.64	25.23	4.34
July	33.12	1.92	32.69	5.43	26.02	4.10
Aug	33.85	2.11	33.11	4.50	25.11	4.15
Sep	33.21	3.82	32.80	5.19	25.01	3.75
Oct	33.20	3.78	32.69	3.85	24.51	2.64
Nov	33.00	4.51	32.59	5.19	24.29	4.43
Dec	33.14	3.97	32.62	5.79	24.21	4.32

Table 9
Economic characteristics of optimized grid/diesel systems.

Location	System	Grid (kW)	Diesel (kW)	NPC (\$)	COE (\$/kWh)	RF
Hawassa IP	grid/diesel	80	50	102,290	0.042	0
Kombolcha IP	grid/diesel	80	50	116,123	0.048	0
Adama IP	grid/diesel	80	48	73,511	0.047	0

Table 10
Annual emissions (kg/yr) from diesel generator during grid/diesel systems operation.

Location	CO ₂	CO	NO ₂	SO ₂	PM	Unburned Hydrocarbon	Total Emissions (kg/yr)
Hawassa IP	1,155	8.13	7.05	2.95	0.05	0.38	1,174
Kombolcha IP	4,394	27.21	25.31	10.81	0.16	1.21	4,459
Adama IP	2,495	16.1	14.91	6.11	0.09	0.69	2,533

Table 11
Annual electric production and consumption of the three regions.

Locations	Grid Purchase (kWh/year)	Diesel generation (kWh/year)	Load (kWh/year)
Hawassa IP	118,998	1,295	119,885
Kombolcha IP	114,365	4,679	118,102
Adama IP	74,168	2,674	76,102

Table 12
Monthly average electric production of grid/diesel/PV/battery systems.

Month	Hawassa IP			Kombolcha IP			Adama IP		
	Grid (kW)	Diesel (kW)	PV (kW)	Grid (kW)	Diesel (kW)	PV (kW)	Grid (kW)	Diesel (kW)	PV (kW)
Jan	29.65	1.45	7.12	28.95	2.98	7.75	22.30	2.50	4.58
Feb	28.05	0.68	8.01	28.48	3.14	6.55	21.98	1.00	4.88
Mar	30.00	1.54	7.14	29.45	2.99	7.65	22.20	1.23	4.60
Apr	30.00	2.13	6.95	29.10	2.89	8.00	21.99	2.40	4.57
May	29.99	1.56	7.01	29.00	3.49	6.78	21.98	2.50	4.53
Jun	30.12	2.13	6.93	29.01	3.41	7.75	22.20	2.71	4.31
Jul	31.11	3.32	6.01	29.12	3.51	6.60	22.35	2.87	4.10
Aug	31.01	3.15	6.10	31.23	3.99	6.20	23.50	3.82	3.70
Sep	29.93	2.98	6.23	29.98	3.89	6.58	22.43	3.12	4.22
Oct	30.00	1.59	7.00	29.42	3.75	6.55	22.23	2.71	4.58
Nov	29.87	1.56	7.11	28.99	3.84	6.53	21.67	2.79	4.57
Dec	29.78	1.41	7.17	28.89	2.65	7.51	22.24	2.74	4.30

wind first into mechanical power, and the generator converts the mechanical power into electric power. The actual power available in the wind turbine is given by Eq. (9) (Mathew, 2006; Wagner and Mathur, 2009).

$$P = \frac{1}{2} \rho A_T V^3 C_p \quad (9)$$

Where P is the output power (W), ρ is the density of air (kg/m^3), A_T is the swept area (m^2), V is the wind speed (m/s^2), and C_p is the power coefficient. To check the viability of the site for wind power production, 10 kW wind turbine is selected. The power-speed characteristic curves and the technical specifications of the selected turbine are given in Fig. 18 and Table 4, respectively.

3.4.4. Battery

Battery stores direct current (DC) electrical energy in electro-chemical form for supporting distributed generators. In addition to supporting the renewable resource, the battery Storage system can improve the grid power quality (Bahmani-Firouzi and Azizpanah-Abarghoee, 2014). Surrette 6CS25P battery model that are appropriate for renewable systems were chosen for this study. The minimum a state of charge point for the batteries has been set at (40%). When this point is reached, the battery does not discharge to avoid the damage with excessive discharge. The discharging and charging equation of the battery is given by Eqs. (10) and (11) (Haidar et al., 2011; Thomas et al., 2016).

For battery discharging

$$E_{bt}(t) = E_b(t-1)(1-\sigma) - [E_{bh}(t)/n_{bi} - E_{bl}(t)] \quad (10)$$

For battery charging

$$E(t) = E_b(t-1)(1-\sigma) + [E_{bh}(t) - E_{bl}(t)]n_{bb} \quad (11)$$

Where E_b is the battery energy in time interval, E_{bh} is the total energy generated by PV array, E_{bl} is the load demand in time interval, n_{bi} is the inverter efficiency, n_{bb} is the battery charging efficiency, and σ is the self-discharging factor. The technical specifications of the selected battery are listed in Table 5.

3.4.5. Power convertor

A converter is used to convert electric power from DC to AC in a process called inversion and from AC to DC in a process called rectification. Both conversion processes are important. The efficiency of inverter and rectifier is taken as 95% (Duman and Güler, 2018; Gökçek, 2018). The specifications of the selected converter for this study are given in Table 6.

3.4.6. Diesel generator

A diesel generator is an ancillary component of the existing supply system, and it consumes diesel fuel to produce electricity. The fuel consumption of the generator as a function of its electrical power output is given by the following equation (Diab et al., 2016; Dufo-lo, 2008).

$$F = F_0 Y_{gen} + F_1 P_{gen} \quad (12)$$

Where F is fuel consumption (L/hour), F_0 is generator fuel curve intercept coefficient (L/hr/KW_{rated}), F_1 is the fuel curve slope in L/hr/kW_{output} , Y_{gen} is the rated capacity of the generator in kW, and P_{gen} is the electrical output of the generator in kW. The technical data of the existing diesel generator operating in the industrial parks are presented in Table 7. The capital cost is not considered because the generators have been already installed.

3.5. System economics

Once the resource and the component data are entered, HOMER Pro ranks all optimization results according to their total net present costs

Table 13
Economic characteristics of optimized grid/diesel/PV/battery systems.

Location	System	Grid (kW)	Diesel (kW)	PV (kW)	Battery	Converter (kW)	NPC (\$)	COE (\$/kWh)	RF
Hawassa IP	grid/diesel/PV/battery	80	50	10	4	6	107,855	0.044	13.3
Kombolcha IP	grid/diesel/PV/battery	80	50	10	6	6	116,998	0.049	13.4
Adama IP	grid/diesel/PV/battery	80	48	6	3	4	74,044	0.048	12.9

Table 14
Annual emissions (kg/yr) from diesel generator for grid /diesel/PV/battery systems.

Location	CO ₂	CO	NO ₂	SO ₂	PM	Unburned Hydrocarbon	Total Emissions (kg/yr)
Hawassa IP	640	4.53	4.12	1.89	0.038	0.247	651
Kombolcha IP	2,444	15.6	14.3	5.98	0.092	0.672	2,475
Adama IP	1,449	9.36	8.68	3.55	0.055	0.399	1,471

(NPC). The NPC is analyzed with the help of the following equation (Duman and Güler, 2018; Haratian et al., 2018; Sinha and Chandel, 2015):

$$NPC = \frac{C_T}{CRF(i, N)} \quad (13)$$

Where C_T is the total annualized cost (\$/year), i is the annual real interest rate (%), N is the lifetime of the project (year), which taken as 25 years for the study, and CRF is the capital recovery factor. CRF is calculated using the following equation (Phurailatpam et al., 2018; Singh et al., 2017):

$$CRF(i, N) = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (14)$$

The annual real interest rate is calculated from the nominal interest rate and inflation rate using the following equation (Dursun, 2012; Ramli et al., 2015):

$$i = \frac{i_n - f}{1 + f} \quad (15)$$

Where i_n is the nominal interest rate (%), and f is the annual inflation rate (%). In this study, a 12.75% nominal interest rate and a 10.84% annual inflation rate is taken (National Bank of Ethiopia, 2018). With these values, using Eq. (15), the real interest rate is found to be 1.72%.

The levelized cost of energy (COE) is the per kWh cost of the power plant over an assumed duty cycle and, is calculated as follows (Haghighat Mamaghani et al., 2016; Zou et al., 2017):

$$COE = \frac{C_T}{E_p + E_d + E_{gs}} \quad (16)$$

Where E_p is the total amount of primary load that the system serves per year (kWh/yr), E_d is the total amounts of deferrable load that the system serves per year (kWh/yr), and E_{gs} is the amount of energy sold to the grid per year (kWh/yr).

Table 15
Annual electric production and consumption of the three regions.

Locations	Grid Purchase (kWh/year)	Diesel generation (kWh/year)	PV Generation (kWh/year)	Load (kWh/year)	Grid sales (kWh/year)	Total consumption (kWh/year)
Hawassa IP	102,890	698	18,041	119,885	35.78	119,921
Kombolcha IP	99,112	2,634	17,405	118,102	12.5	118,115
Adama IP	64,768	1,545	10,864	76,102	26	76,128

4. Simulation result and discussion

Based on the available energy resources and the load demand, HOMER Pro simulates every feasible power supply configuration. In the following subsections, the techno-economic and environmental feasibility analysis of grid/diesel, grid/diesel//PV/battery, grid/PV/battery, and grid/diesel/wind/PV/battery power systems configuration to supply the industrial park load in three different regions of Ethiopia were simulated. Optimization and sensitivity analysis are the two basic simulation results that can be performed by HOMER Pro.

4.1. Optimization results

4.1.1. Grid/diesel generator systems

Grid/diesel generator systems were simulated and analyzed for comparison purposes. In this scenario, diesel generator is fully accountable for supplying the critical loads of the parks during a grid outage. The average monthly electric production from grid/diesel systems for the study regions are shown in Table 8.

Economic and environmental considered output of the grid/diesel systems for the investigated locations are shown in Tables 9 and 10. Without considering the capital and replacement costs, the NPC and COE of the three regions of grid/diesel systems were \$102,290 and 0.042 \$/kWh (Hawassa IP), \$116,123 and 0.048 \$/kWh (Kombolcha IP) and \$73,511 and 0.047 \$/kWh (Adama IP). These systems produce a total of 1174 kg/yr, 4459 kg/yr, and 2533 kg/yr gas emission in Hawassa, Kombolcha, and Adama, respectively.

The annual energy production from grid/diesel systems and annual consumption for the three regions are shown in Table 11.

4.1.2. Grid/diesel/PV/battery systems

In this system architecture, PV/battery systems help the diesel generator during a grid outage. The monthly average electric productions from grid/diesel/PV/battery systems for the three regions are shown in Table 12. The minimum and maximum monthly power production from the grid and diesel are also shown in this table. In general when we observe the three locations, the PV systems were capable enough to produce a power throughout the year in all three climate zones.

The economic and annual emission characteristics of the optimized grid/diesel/PV/battery systems for the study regions are shown in Tables 13 and 14. Because of the highest grid outage and highest consumption from diesel generator in Kombolcha IP, the NPC (\$116,998) of the systems was also the highest among the three regions. The lowest NPC (\$74,044) was observed in Adama IP. Lowest emissions

Table 16
Economic characteristics of optimized grid/PV/battery systems.

Location	System	Grid (kW)	PV (kW)	Battery	Converter (kW)	NPC (\$)	COE (\$/kWh)	RF
Hawassa IP	grid/PV/battery	80	30	20	25	136,151	0.056	41.6
Kombolcha IP	grid/PV/battery	80	35	25	35	151,432	0.060	48.2
Adama IP	grid//PV/battery	80	25	20	25	104,120	0.063	49.6

Table 17
Economic output of optimized grid/diesel/PV/wind/battery systems.

Location	Systems	Grid (kW)	Diesel (kW)	PV (kW)	Wind (kW)	Battery	Converter (kW)	COE (\$/kWh)	RF
Adama IP	grid/diesel/wind/PV/battery	80	48	6	10	3	4	0.081	22.1

Table 18
Annual emissions (kg/yr) from diesel generator for grid/diesel/PV/wind/battery systems.

Location	CO ₂	CO	NO ₂	SO ₂	PM	Unburned Hydrocarbon	Total Emissions (kg/yr)
Adama IP	1,270	8.50	7.94	3.20	0.046	0.320	1,290

were observed in Hawassa IP due to the lowest grid outage consideration. The highest emissions were found in Kombolcha IP.

The summery of annual energy production from grid/diesel/PV/battery systems and annual consumption for the three regions are shown in Table 15.

4.1.3. Grid/PV/battery systems

Among the common possible combination in three regions, the NPC and COE of grid/PV/battery systems were the highest. However, there is no emission in this configuration. The economic characteristics of the systems are shown in Table 16.

4.1.4. Grid/diesel/wind/PV/battery systems

From the three regions, wind power was technically and environmentally feasible only in Adama IP. However, the COE for wind integrated power systems was considered as the highest among all possible configurations for this region. The economic and environmental oriented outputs of the systems are shown in Tables 17 and 18.

4.1.5. Economic and environmental comparative analysis of the proposed four systems

In order to select the best supply option for the locations from the proposed scenario, the economic and environmental-related parameter comparison is essential. The economic and emission parameter that serves as a feasibility determinant factors for each regions are shown in

Table 19. For all three climate regions, the grid/diesel systems were the lowest cost systems; however, the total annual emissions of systems were the highest. Emissions from grid/diesel/PV/battery systems were lower than grid/diesel systems for all regions. Grid /PV/battery systems are emission free systems, but the COE of the grid /PV/battery systems were higher than the COE of grid/diesel/PV/battery systems. The COE of grid /diesel/wind/battery systems were the highest among the considered supply option in the region.

4.2. Sensitivity analysis results

Sensitivity analysis indicates the effect of changing multiple values for a particular input variable on the optimal system design. Such variables can be solar radiation, PV capital cost, grid tariff, load demand, fuel prices, grid outage, and any other related parameters that might affect the economic output of the optimal system. In this investigation, the effects of electric load change, solar radiation, grid tariff, and grid outage on the NPC and COE for the optimal system design of grid/diesel/PV/battery systems were explored.

4.2.1. Effect of load change

As it is stated in Section 3.2 previously, the daily average critical load demands of the sites are considered as 329 kW h/day (Hawassa IP), 321 kWh/day (kombolcha IP), and 208 kW h/day (Adama IP). However, by considering the dynamic nature of the loads, the energy demands of the three regions are varied from 300 kW h/day to 380 kW h/day for Hawassa IP and kombolcha IP and 200 kW h/day to 280 kW h/day for Adama IP. The performance of total NPC and COE has been measured accordingly. As shown in Fig. 19, the NPC of the systems for all three sites increased with increasing load demand, whereas the COE decreased with increasing load demand.

Table 19
A comparison between the four scenarios for the three regions.

Systems	Hawassa IP			Kombolcha IP			Adama IP		
	NPC (\$)	COE (\$/kWh)	Annual emission (kg /yr)	NPC (\$)	COE (\$/kWh)	Annual emission (kg /yr)	NPC (\$)	COE (\$/kWh)	Annual emission (kg /yr)
grid/diesel	102,290	0.042	1,174	116,123	0.048	4,459	73,511	0.047	2,533
grid/diesel/ PV/battery	107,855	0.044	651	116,998	0.049	2,475	74,044	0.048	1,471
grid/PV/ battery	136,151	0.056	0	151,432	0.060	0	104,120	0.063	0
grid/diesel/ wind/PV/ battery	–	–	–	–	–	–	129, 125	0.081	1,290

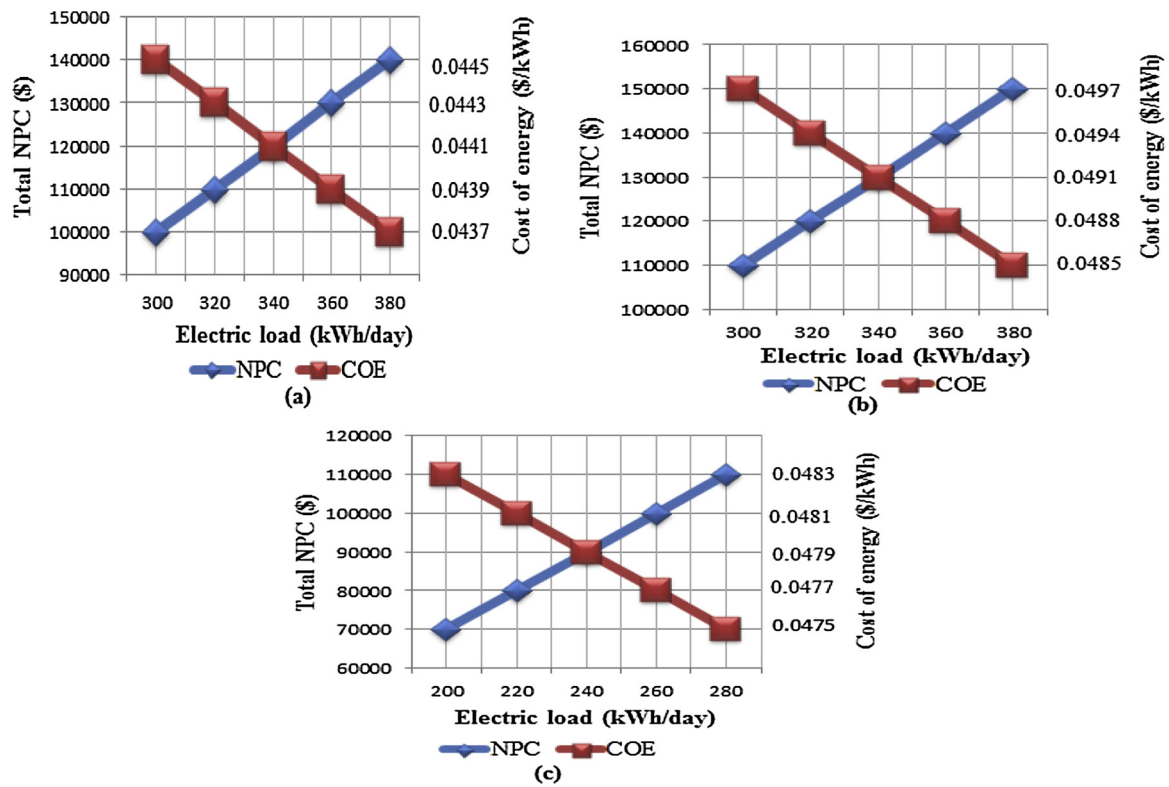


Fig. 19. Effect of electric load change on the NPC and COE for the study regions ((a) Hawassa IP, (b) Kombolcha IP, and (c) Adama IP).

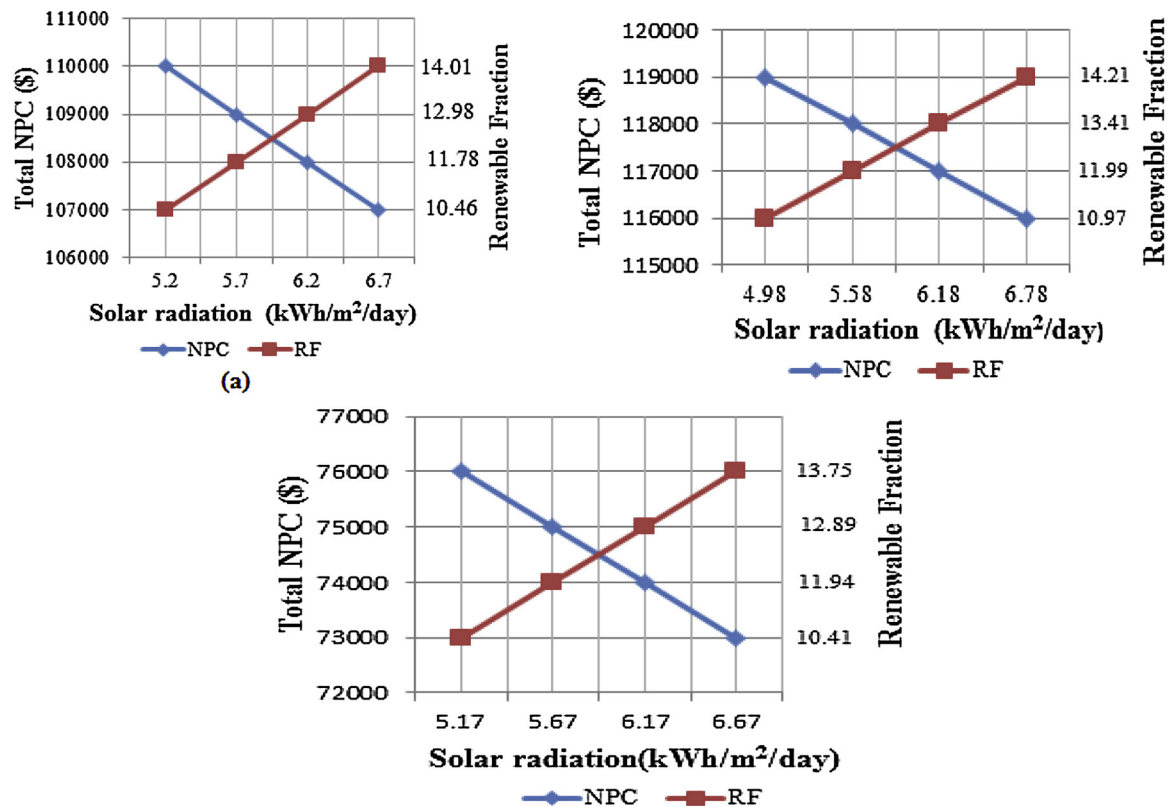


Fig. 20. Effect of solar radiation variation on the NPC and RF for the study regions ((a) Hawassa IP, (b) Kombolcha IP, and (c) Adama IP).

4.2.2. Effect of solar radiation variations

Fig. 20 shows the results of sensitivity analyses of the optimized grid/diesel/PV/battery systems to variations in solar radiation. The load (329 kWh/day, 321 kWh/day and 208 kWh/day for the Hawassa

IP, Kombolcha IP, and Adama IP) and grid power price (0.04 \$/kWh) were all fixed simultaneously. The NPC values of the three locations decreased as global solar radiation values increased, but the RF values of these three locations increased as the solar radiation increased. This

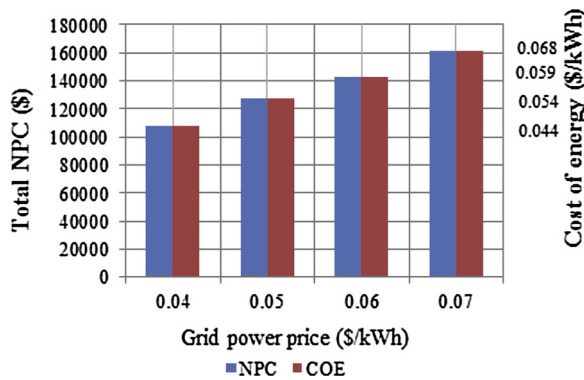


Fig. 21. Effect of grid power price on the NPC and COE for the study region.

suggests that solar PV penetration in the grid is very important for reducing energy-related environmental concerns.

4.2.3. Effect of grid power price variations

The grid tariff is varied from 0.04 \$/kWh to 0.07 \$/kWh for all regions. The NPC and the COE of all locations increased with increasing power prices. For all regions, the rate of grid power price at a country level is the same. Therefore, it is enough to see one region as a sample. Fig. 21 below shows the effect of grid power price variation on the NPC and COE for Hawassa IP.

4.2.4. Effect of grid outage variations

By keeping the total load and the grid power price of each region constant, the line interruption duration of the three regions was varied

from 80 to 110 hr/year for Hawassa IP, 230–260 hr/year for Kombolcha and 160–190 hr/year for Adama IP, respectively. As shown in Fig. 22, the NPC and the COE of all sites increased with increasing line interruption duration. In the analysis, the interruption frequency was varied proportionally with the interruption duration, and the unscheduled outage was considered.

5. Conclusion

In this paper, the techno-economic feasibility analysis of grid-connected PV/wind power systems considering unreliable grid/diesel systems for the industrial park (IP) load in three different regions of Ethiopia was analyzed. NPC, COE, and emissions were considered as basic parameters to indicate the feasibility of the designed systems.

Based on the objective of the study and by considering the feasibility determinant constraints, the main conclusions of the investigation are presented as follows:

- Solar energy was potentially feasible for all three regions. The analyzed annual average solar radiation for Hawassa IP, Kombolcha IP, and Adama IP were 6.18 kWh/m²/day, 5.96 kWh/m²/day, and 6.20 kWh/m²/day respectively. The annual average temperature for the locations were 27.9 °C (Hawassa IP) 27 °C (Kombolcha IP), and 28.14 °C (Adama IP), respectively.
- Except for Adama IP, Hawassa IP and Kombolcha IP locations were not feasible for wind power production. The average annual maximum and minimum wind speed at 50 m for the sites were 4.9 m/s, 3.1 m/s (Hawassa IP), 4.6 m/s, 2.84 m/s (Kombolcha IP)

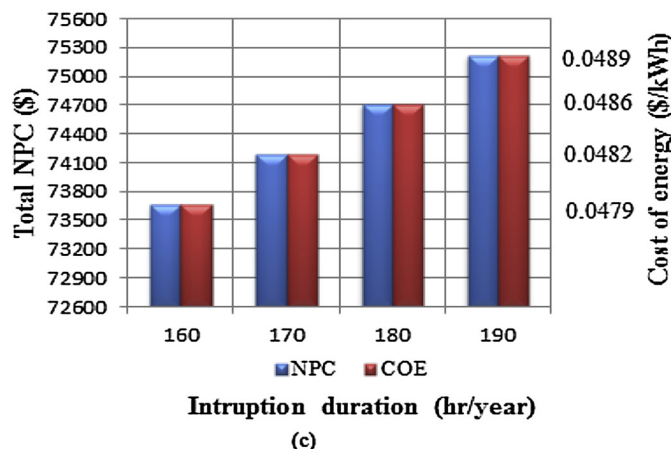
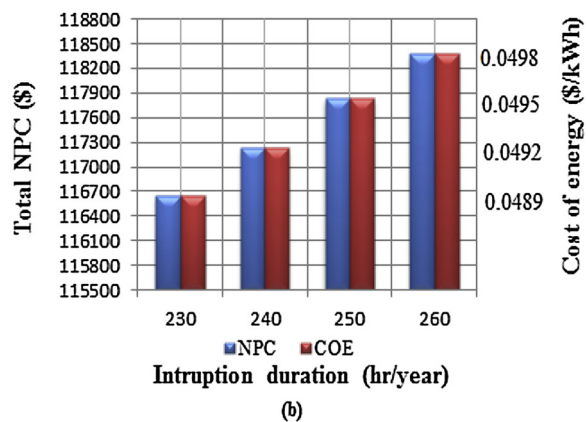
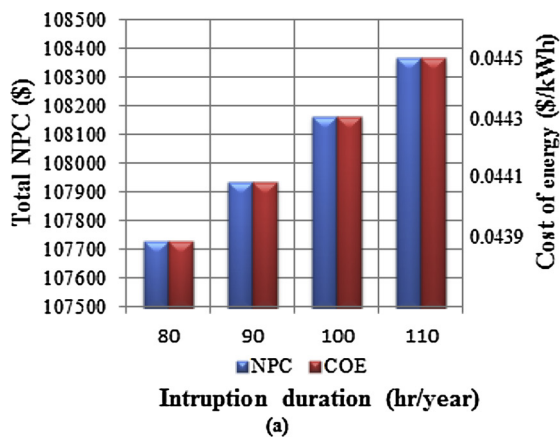


Fig. 22. Effect of grid outage variation on the NPC and COE for the study regions ((a) Hawassa IP, (b) Kombolcha IP, and (c) Adama IP).

and 6.7 m/s, 3.8 m/s (Adama IP) respectively.

- The grid/diesel systems were the lowest cost systems among the other considered configuration for all three regions, but the emissions for grid/diesel systems were relatively the highest.
- In grid/PV/battery systems there are no emissions; however, the COE of the grid /PV/battery systems was higher than the COE of grid/diesel/PV/battery systems.
- Grid/diesel/wind/PV/battery systems were technically feasible in Adama IP, but the COE of the systems was higher than the remaining possible other configurations of the region.
- Based on technical, economic, and environmental consideration, the grid/diesel/PV/battery systems were the best supply option for all three regions among the considered systems.
- From sensitivity analysis, it is observed that the optimal grid/diesel/PV/battery systems operate reasonably well with the variation of electric load, solar radiation, grid power price, and grid outage variation.
- The NPC of the optimal systems for all three regions increased as the electric load increased, and the COE decreased as the electric load increased.
- The NPC and the emission of the optimal systems in all three places decreased as the solar radiation increased
- The NPC and the COE of the optimal systems in all three sites increased as grid tariff increased.
- The NPC and the COE of the optimal systems in all three regions increased as the line interruption duration increased.
- In general, based on the results from this investigation, the implementation of grid-integrated solar PV systems in the south, north-central and central part of Ethiopia could be environmentally and economically viable. Thus, the finding of this research study will provide a convincing pathway to the concerned stakeholders and policy makers to extend these systems into the aforementioned regions for application to other firms also.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. In short, the authors declare they have no conflict of interest.

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